

the URLLC will help in navigation and traffic control.

Green Revolution 2.0

Real-time fast data transition, IoT connected devices and sensors, video-controlled irrigation, and nutrients and pesticides control systems will enable the big farmers to monitor health of their crops and livestock and continuously improve the irrigation efficiency to reap higher yields. Small farmers will be able to improve irrigation efficiency and crop yield using virtual training programs.

Using a range of sensors, the renewable energy farms will be able to continuously monitor the factors impacting the health of the solar panels and windmills, thus improving energy production. The response time and efficiency of the wind and solar renewable energy farms will get radically improved using the fast speed low latency networks.

Bedrock of Industry 4.0

Using an array of IoT sensors and devices, and automating the scheduling of processes, 5G networks will become the cornerstone of Industry 4.0 to improve efficiency and reduce the carbon emissions. Work stoppages will get minimised by automating the just-in-time supply chains and implementing predictive maintenance.

It will be possible to make



Industry 4.0—a fusion of several related technologies (Source: <https://dat4zero.eu>)

available a number of B2B services, such as wirelessly controlled robots in factories allowing every component—even the smallest—to be wirelessly tracked, operated, and managed. Use of robot to robot and worker to robot communication will help achieve higher output and efficiency.

The sales and marketing will be able to provide novel immersive experiences in trying, styling, and buying products online as well as in brick-and-mortar stores.

Advanced facilities for smart cities

Advanced facilities, such as voice and gesture remote controlled houses, improved waste management systems, and shopping malls, that provide exciting audio-visual experiences, will become available in smart cities. Metaverse and digital applications will provide extraordinary immersive experience

to spectators in stadiums, open air festivals, and other public events.

Automatic traffic lights and communication channels, such as machine-to-machine (M2M), vehicle-to-everything (V2X), device-

to-device (D2D), and human-to-machine will make possible advanced vehicle tracking, wall-mounted sensors, and improved utility services.

5G work in progress

Twenty-five years ago, it would have been virtually impossible to predict the frantic pace of progress the telecom sector in India has made. No longer limited to the basic phone and internet services, the sector is now playing a pivotal role in providing multifunctional mobile communications, social connectivity, and fast internet services. The telecom sector has become the driving engine for digitisation and a catalyst to spur economic growth and social progress.

Telecom operators are working 24/7 to roll out 5G services pan India. 5G, the fastest-adopted mobile technology, is set to unlock multiple untapped opportunities, unleashing faster levels of connectivity, changing forever the way people interact and conduct businesses, generating employment opportunities, directly and indirectly contributing to economic and financial development and the way governments govern. Remote controlled smart houses; Industry 4.0 processes, machine-to-machine communications and predictive maintenance; application of robotic process automation (RPA); and cloud computing is expected to create a huge social, educational, and economic impact affecting how people live, work, communicate, en-



5G enabled IoT is revolutionising transportation and smart city traffic management (Source: <https://www.digi.com>)